Enhancing Cloud Interoperability and Usability in Bioinformatics

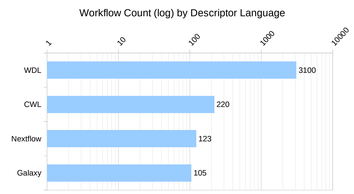
David Steinberg[1], Denis Yuen[2], Charles Overbeck[1], Kathy Tran[2], Steve Von Worley[1], Nayeon Hyun[2], Ben Vizzier[1], Lucia Zhang[2], Benedict Paten[1], Lincoln Stein[2]
[1] University of California, Santa Cruz Genomics Institute, Santa Cruz, CA USA. Email: davidcs@ucsc.edu
[2] Ontario Institute for Cancer Research, Toronto, ON Canada.

Background

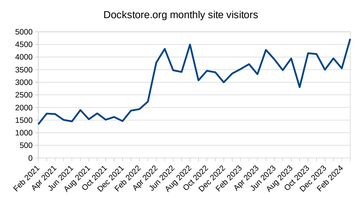
Dockstore is a free and open source platform for sharing scientific tools and workflows available at **dockstore.org**. It is primarily a registry of Docker-based resources described using popular workflow languages in bioinformatics such as CWL, WDL, Nextflow, Galaxy, and now Jupyter notebooks.

Our initial aims were to increase portability (sharing workflows that can run in any environment supporting Docker), interoperability (championing and implementing GA4GH APIs), and reproducibility (preserving dependencies via Docker and describing workflows with popular descriptor languages). These efforts align closely with the FAIR methodology.

The current Dockstore version is 1.15.2. Dockstore versions 1.11 and 1.6.0 were presented at BOSC 2021 and 2019 respectively.



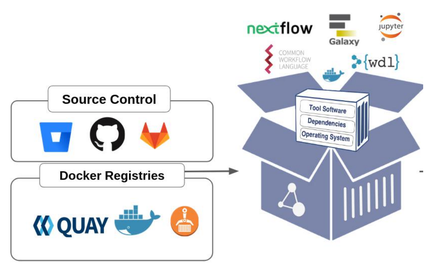
Public API Access
100+ GitHub Stars
2500+ workflows
100+ total citations



Integrations and Key Features

Source control and container registry integration

Dockstore encourages developers to use source control in the development of their pipelines by offering integrations with GitHub, Gitlab, and Bitbucket. This allows any changes to underlying source control management systems to be reflected in Dockstore including versions, tags, and branches of git repositories. Dockstore allows pipeline developers to use the Docker registry of their choice and encourages use of publicly accessible containers.



Digital Object Identifiers and ORCiD

As demonstrated in previous BOSC sessions, Dockstore assists with generating Digital Object Identifiers (DOIs) for pipelines using linked Zenodo accounts. Open Research and Contributor ID (ORCID) integration further allows authors of pipelines to be correctly identified and their efforts to appear as works in the ORCID network.

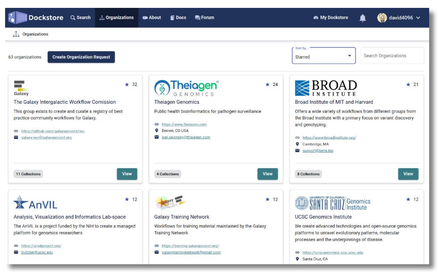
DOI 10.5281/zenodo.7259426

ORCID Account
https://orcid.org/0000-0001-6683-2270

Organizations and Collections

Organizations such as labs and academic institutions are able to make their works more findable and accessible by creating a presence on Dockstore. This allows verified organization members to curate their pipelines and highlight specific versions as part of Dockstore collections. Verified organizations include public efforts such as public health labs, the Galaxy Intergalactic Workflow Commission, the Human Pangenome Reference Consortium, and the ENCODE project as well as private labs like PacBio and Theiagen Genomics.

60+ Organizations



Cloud Launch-with platforms

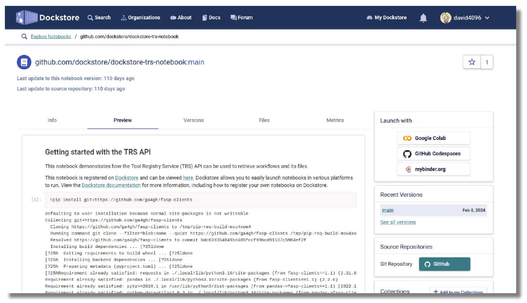
To fulfill the aim of encouraging reusability and interoperability for workflows Dockstore integrates with a variety of cloud execution systems for pipelines. Execution platforms interact with Dockstore via the Global Alliance for Genomics and Health (GA4GH) Tool Registry Service (TRS) API, which Dockstore implements. Execution platforms are encouraged to add their integration to the Dockstore system by becoming platform partners allowing users to easily launch workflows in the cloud.

12,000+ workflows launched

Jupyter Notebook Integration

Jupyter notebooks are a growing resource for analysis, generation of diagrams, and sharing pipeline results. To enable researchers to share their work Dockstore has integrated Jupyter notebooks as a first-class citizen of the system.

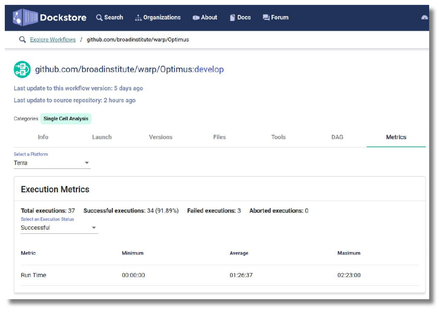
The benefits of adding Jupyter notebooks to Dockstore are similar to those of workflows. They can be discovered via search, previewed for their content, provided DOIs, curated in collections, and launched into execution environments. The Launch-with environments for notebooks include Google Colab, GitHub Codespaces, and mybinder.org.



With the addition of this feature Dockstore hopes to encourage not only the findability and accessibility of notebooks using search and DOIs but also their reusability and portability by offering a way to provide configuration files needed to recreate the execution environment.

Execution Metrics and API

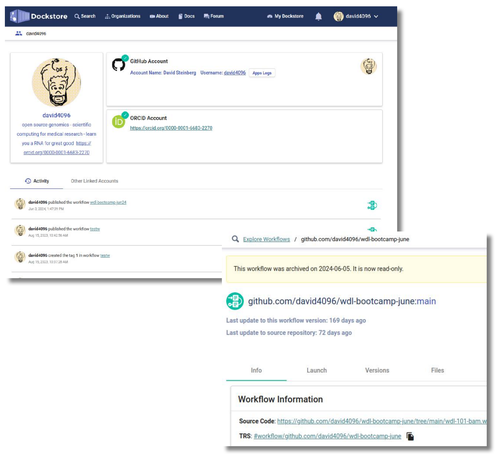
One barrier to reuse of pipelines is the unpredictable nature of the costs related to their use. Dockstore aims to assist in the reuse and sharing of pipelines by providing a data model and API for platform partners to share metrics related to a workflow run. These include status reports that allow Dockstore to show whether a workflow ran successfully, as well as runtime information that will allow researchers to estimate the cost of reusing a given workflow. As Dockstore displays metrics for specific workflow versions, developers can use these data to compare how a workflow has changed over the course of its development.



As Dockstore itself is not an execution platform we are actively seeking platform partners to provide metrics using this open data model. Future developments may include visualization and other tools for cost estimation.

User profiles

Dockstore is committed to making it as easy as possible to follow the FAIR methodology for researchers authoring bioinformatics pipelines. The system now displays user profiles and each profile is given its own unique URL. This provides a public facing profile where ORCiD identifiers and linked accounts are displayed. User activity on the site is displayed on profiles and made accessible to other Dockstore users.



Archiving

In addition, Dockstore hopes to address the long term maintenance concerns of pipeline developers by providing features for every part of a pipeline's development lifecycle. By enabling them to archive work that is no longer meant to be updated developers can signal to other researchers that a pipeline is no longer actively maintained.

Future developments

- Simplified DOI creation
- Improved execution metrics visualization
- AI topic sentence generation

Visit **dockstore.org** to learn more

Supported by
AnVIL SOW (Grant #2U24HG010262-06)
ODSP Africa SOW (Grant #5U2CEB032224-03)
Dockstore U24 (Grant #5U24HG011853-04)
Genome Canada
Government of Ontario

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